

SIRATCHA BOONSAKDEE

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PROFESSIONAL SUMMARY

Engineering student who enjoys real conversation across industries to understand people's needs, expertise, and lifestyles — using those insights to find gaps in existing solutions and improve how they're delivered and adopted. Curious by nature, especially drawn to the logic behind innovation in smart systems and semiconductors.

EDUCATION

Singapore University of Technology and Design (SUTD)

Bachelor of Engineering (Engineering Product Development)

Specialization: Electrical Engineering | Minors: Psychology & Business Management

- Scholarship: Kewalram Chanrai Group Scholarship for Women | GPA: 4.33 / 5.00
- Scholarship: Scholarships and Educational Exchanges for Development – Phase 2 (SEED-2) program - Canada

Singapore

16 Sep 2024 – 29 Apr 2028

Kamnoetvidya Science Academy (KVIS)

- *High School Diploma Full Scholarship (3 Years) | GPA: 3.86 / 4.00*

Thailand

May 2021 - Mar 2024

SKILLS

- **Engineering:** Circuit Design & Prototyping, PCB Assembly & Soldering, KLayout, CST Studio Suite, KiCad, Photolithography, Embedded Systems (Arduino & Raspberry Pi), Python, Basic C, MATLAB, Design of Experiments (DOE), Fusion 360 (CAD), 3D Printing
- **Languages:** Thai (Native), English (Proficient), Chinese (Beginner)

WORK EXPERIENCE

Chulalongkorn University

Research Internship

Bangkok, Thailand

Apr 2023 – May 2023

- Synthesized AuNPs and optimized colorimetric pesticide sensors using DOE varying solution volume, concentration, and pH; analyzed SEM, TEM, and EDS data to identify morphology–performance relationships, controlled lab testing and resolved detection sensitivity variances, earning a Regional Silver Medal at the NSTDA Young Scientist Competition.

ACADEMIC RESEARCH PROJECTS

- **SUTD – Retail Shelf Monitoring Robot** **May 2026 – Present**
 - Evaluated problems across various industries and selected retail shelf monitoring as the final focus area, aligning with Koryo Mart's business needs, to propose a robotics-based solution.
 - Integrated a LiDAR sensor supporting SLAM, ROS, and Nav2-based navigation, and contributed to power management, moving base construction, and PCB design.
- **SUTD – Diffraction Grating Design** **May 2026 – Present**

Designed and fabricated a diffractive-grating souvenir on silicon in KLayout and patterning it by optimising photoresist spin-coating, direct-laser-write exposure, and development parameters through a DOE approach to render the structural colour achieving achieving a 1 μm line width at 33% duty cycle (3 μm pitch).
- **SUTD – DynaGuard Smart Knee Brace System** **Mar 2026 – May 2026**

Designed the electronics system for DynaGuard, an adaptable smart knee brace, with a team of 4. Focused on embedded system architecture, compact component placement and closed-loop pressure control, iterating on the hardware to achieve consistent strap adjustment within 2–3 seconds.
- **SUTD – Solar-Powered In-Car Air Purifier System** **Jun 2025 – Aug 2025**

Engineered a solar-powered in-car air purifier by designing a power circuit integrating a solar panel, solar charge controller, Li-ion battery, and DC fan motor, applying photovoltaic IV curve analysis and battery sizing principles to ensure reliable energy storage and load management; modelled the enclosure using CAD 3D and validated closed-loop energy harvesting through reverse engineering and data-driven testing of the prototype.

LEADERSHIP EXPERIENCE

- **President, SUTD Beauty of Thai Interest Group** **May 2025 – Mar 2026**

Led operations and events promoting Thai culture, engaging 50+ participants.
- **Welfare Director & Publication Head, SUTD Ambassador Club** **May 2025 – Mar 2026**

Managed club communications and led branding and presentation design for recurring external outreach and school engagement initiatives; delivered public speaking presentations to 100+ total participants across sessions.

EXTRACURRICULAR ACTIVITIES

- **IC Design Summer Camp 2026** **July 2026**

Gained exposure to analog/digital IC design at AMD and Marvell, and explored semiconductor manufacturing processes at STMicroelectronics.
- **SEMI SEA Tech Zoomer Regional Bootcamp 2026** **May 2026**

Served as Discussion Lead in a bootcamp project on data-driven packaging reliability risk prediction using manufacturing process data under Micron mentorship, with exposure to semiconductor industry workflows through industry talks and a site visit to MIMOS Berhad.
- **SUTD Translating Innovative Intelligent Design to Entrepreneurship Programme** **Sep 2025 – Dec 2025**
 - Selected for a 3-month exchange at Zhejiang University; studied AI tools applications and digital business strategies through direct training from Alibaba and leading Chinese tech firms, supported by a SGD 4,000 SUTD scholarship.
 - Completed a Business Development internship at WorkX, conducting customer segmentation and market analysis for an AI data annotation startup.